

TrES-2b

R-2384

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_120601 · created 2026-08-20T22:48:50+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456079.89214 +/- 0.00015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.0087
Transit depth (Rp/Rs)^2	2.22 +/- 0.26 [%]
Orbital Inclination (inc)	83.71 +/- 0.24
Airmass coefficient 1 (a1)	0.9598 +/- 0.0091
Airmass coefficient 2 (a2)	0.039 +/- 0.043
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	#2 - [70, 136]
Optimal Aperture	4.06
Optimal Annulus	9.69
Transit Duration (day)	0.0755 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.149 $\pm$ 0.0087 vs 0.1254 (deviation 2.42 σ)
Duration fit vs published	0.0755 d vs 0.0746 d (deviation 0.19 σ)
Mid-time O-C	-0.4 min
Depth SNR	15.3
Residual scatter	0.94 %

Light curve

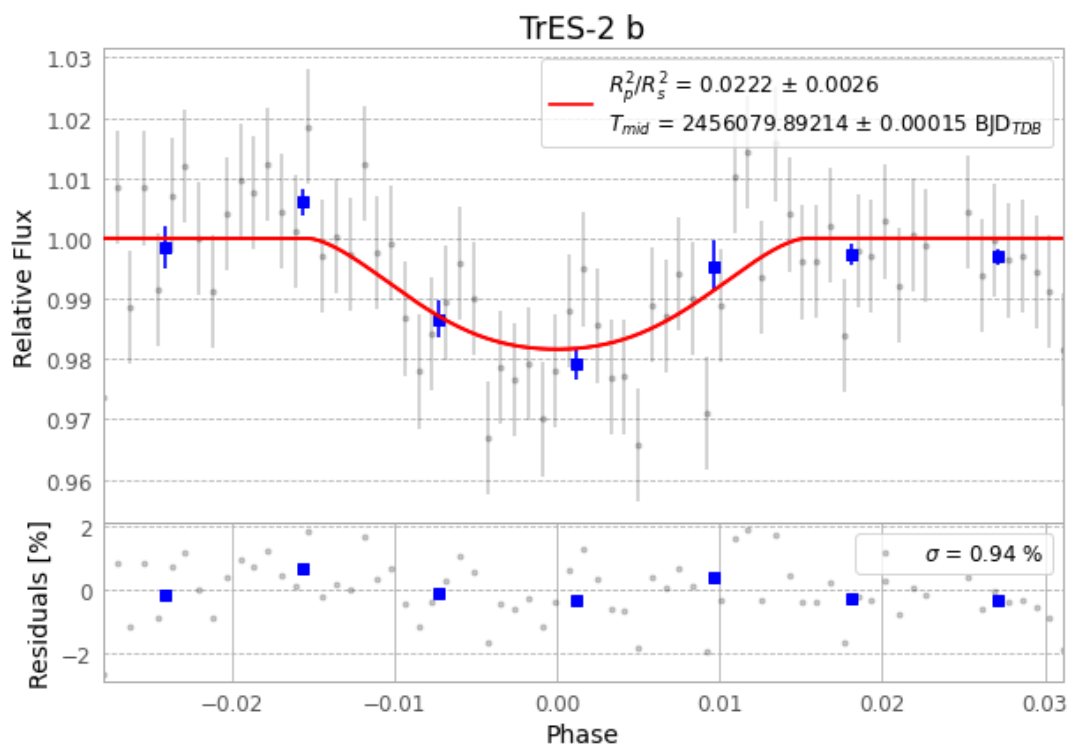


Figure 1 — FinalLightCurve_TrES-2 b_2012-06-01.png (SHA-256 0c2e029a1f3ccd93...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 216], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5098c6e939ac3438...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

71 FITS frames (47 MB), TRES-2120601074212.FITS → TRES-2120601111213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-120601.log (afa5b4df9510...), FinalLightCurve_TrES-2 b_2012-06-01.png (0c2e029a1f3c...), FinalLightCurve_TrES-2 b_2012-06-01.csv (c3336ea20524...)

Compute resources

Wall time 133.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 22:48Z.